

Case Study 9

Pellagra (A Corny Story)

When European explorers returned from North America in the mid 1700s, they brought with them corn. Corn, soon after, became a popular crop for European landowners who would often pay farm workers with a share of the crop yield. Corn thus became an important part of the diet of both North American and European farm workers.



By the late 1700s, farm workers became easy to recognize in a crowd as they often had **crusty reddened skin**. This skin condition was called *pellagra* -- the Italian word for rough skin. Many people believed that pellagra was caused by the consumption of rotten corn. Other symptoms of pellagra included a red tongue, sore mouth, diarrhea, and dementia.

In the early 1900s, pellagra was common in the Southern part of the United States. Many people began to think that pellagra was not caused by spoiled corn but from a communicable infection passed from one person to another. Many of the affected individuals were eventually placed in insane asylums because of the dementia that ensued. Because so many affected individuals were in one location, a physician named Goldberger was able to collect data on how pellagra might be transmitted. What he found from his research indicated that pellagra was not a communicable disease because none of the doctors, nurses, or attendants who worked in the asylum contracted the unique condition.

What then was the cause of pellagra? Goldberger suspected that corn in the diet, or more specifically the overall diet of an individual, played an important role in the development of the disease.

To test his theory that a poor diet caused pellagra, Goldberger asked for and received government funding to supplement the asylum patients' diets with meat and dairy products. To his delight, almost all of the patients recovered after their diets were supplemented!

To prove conclusively that a poor diet caused pellagra, however, Goldberger knew he would have to conduct an experiment to see if a poor diet could induce the pellagra. The question of who to use as test subjects posed a problem for Goldberger. The ethics of the day required willing subjects. Who would volunteer to contract such an affliction?

Goldberger asked the director of a prison in Mississippi if he could use prison volunteers for a six-month study on diet as it relates to pellagra. The director gave approval and inmates who agreed to full participation for the six months would be released at the end of the study. Goldberger had plenty of volunteers!

Participants could eat as much food as they wanted during the six months, but they could eat only corn biscuits, corn mush, corn bread, collard greens, and coffee. After about five months, the prisoners were not very happy. They developed skin lesions, stomachaches, and red tongues. Goldberger had proven conclusively that diet caused pellagra. The specific nutrient that caused pellagra was identified after Goldberger's death. Corn has low levels of niacin, a B vitamin, and a deficiency in this vitamin is what causes pellagra — not rotten corn!